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import cv2 # Import OpenCV for image processing algorithms
import numpy as np # Import Numpy for mathematical matrix operations
import matplotlib.pyplot as plt # Import Matplotlib to visualize the output on screen

# Step 1: Create a high-res image
img = np.zeros((100, 100), dtype=np.uint8) # Create a blank black image array filled with zeros
# Step 2: Draw a diagonal line
cv2.line(img, (10, 10), (90, 90), 255, 2) # Draw a straight line on the image
# Step 3: Simulate poor sampling (downsizing)
low_res = cv2.resize(img, (10, 10)) # Resize image to new dimensions using interpolation
# Step 4: Upscale to show the jagged artifacts
artifact_img = cv2.resize(low_res, (100, 100), interpolation=cv2.INTER_NEAREST) # Resize image to new dimensions using
# Step 5: Display result
plt.imshow(artifact_img, cmap='gray') # Render the image array to the display
plt.title('Sampling Artifacts') # Add a descriptive title to the plot
plt.show() # Show the final plot window to the user
```

Sampling Artifacts

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